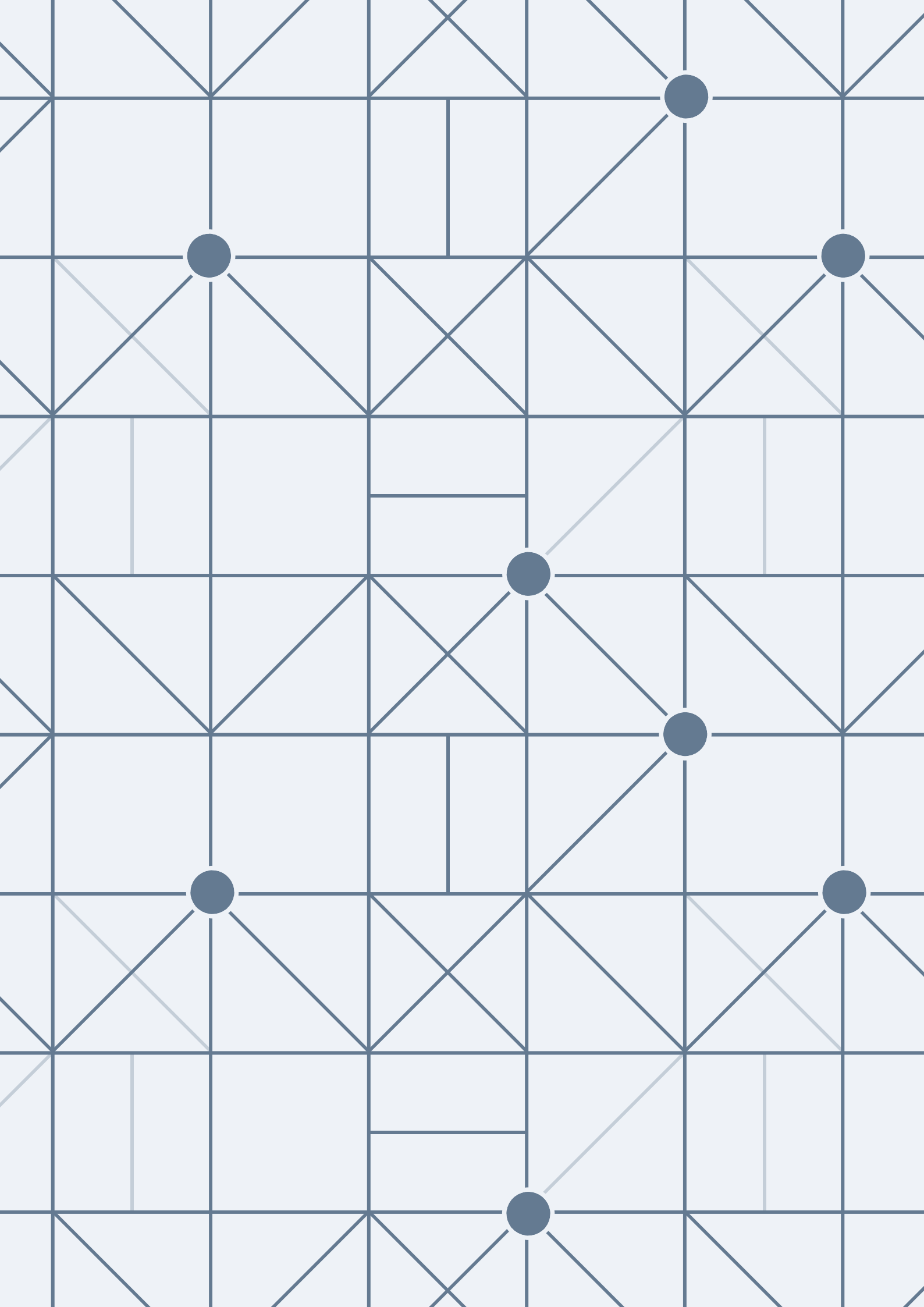
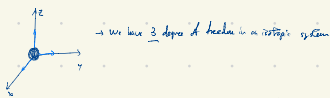
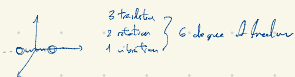
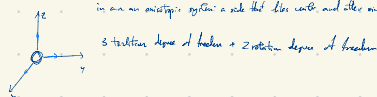




Key words: phase space, degree of freedom



in an anisotropic system a rule that does work and other rule that does not



Not for isotropic: $D(\text{Degrees of freedom}) \rightarrow 5$

D. Freedom	A	Low velocity	Non-bound velocity
Translation	3	3	3
Rotation	0	2	3
Vibration	0	3N-5	3N-6
Total	3N	3N	3N

$\rightarrow$ Always 3N

Phase space encompasses the information of where the particle and where it will be

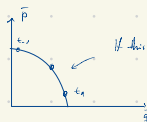
Γ -space (phase space) has 2S dimension ($\vec{q}, \vec{p}$ for each s)

$\vec{p}$ is associated to $\vec{q}$

for example:

$$x, p_x = m \dot{x} \quad 2, p_x = m \dot{x} \quad 0, p_x$$

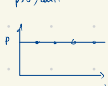
$$y, p_y = m \dot{y} \quad 4, p_x \quad 1, p_x$$



All the particles of my system will be in the same curve and not other place

We only have a simple sphere

$$x(t) = x(0) + \frac{p}{m} t$$



$$p_0 = kx$$

$$m\dot{x} = kx$$

$$\dot{x} = \frac{k}{m}x = 0 \quad u = \sqrt{\frac{k}{m}}, \quad (u) = s^{-1}$$

$$x + u^2 x = 0$$

$$\begin{cases} x = e^{i\omega t} \\ \dot{x} = i\omega e^{i\omega t} \\ \ddot{x} = -\omega^2 e^{i\omega t} \end{cases} \rightarrow \ddot{x} + u^2 x = 0 \rightarrow \ddot{x} + \omega^2 x = 0 \rightarrow \omega = u$$

$$x(t) = C_1 \cos(\omega t) + C_2 \sin(\omega t)$$

$$p = m\dot{x}(t) = -mC_1 \omega \sin(\omega t) + mC_2 \omega \cos(\omega t)$$